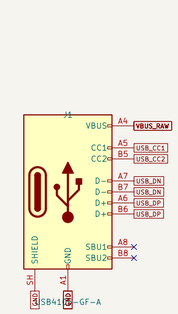
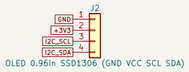
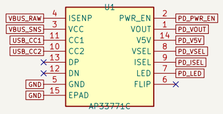
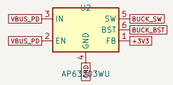
USB-C PD input (AP33771C asks for 9 V), 3.3 V buck, ESP32-C3, OLED + 3 buttons, I2C mux to the six slots.
Label-style schematic generated from design.py: pins with the same label are connected.

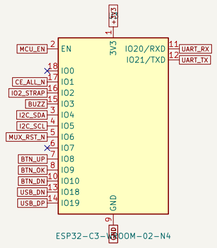






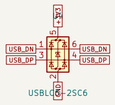


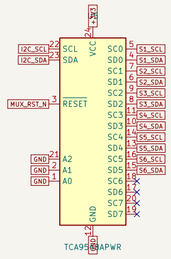


















Sheet: /lipo-charger/
File: lipo-charger.kicad_sch

Title: 6-slot 1S/2S USB-C LiPo charger — power input, MCU, UI

Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.8		Id: 1/14

Slot 1 · Slot 2 · Slot 3 · Slot 4 · Slot 5 · Slot 6

BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

J101
XT30 2S

J102
PicoBlade 1S

J103
XH balance 2S

U102
BQ29209

U101
BQ25798

Q101
AO3401A

Q102
AO3401A

Q103
AO3401A

Q104
AO3401A

Q105
2N7002

Q106
2N7002

Q107
2N7002

L101
2.2µH 3A

D101
CHARGING (green)

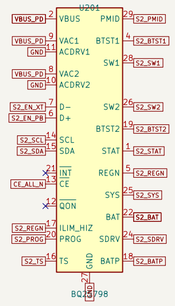
C101 47µF 50V
C102 4.7µF 16V
C103 47µF 50V
C104 1nF 50V
C105 10µF 25V
C106 100nF 50V
C107 10µF 25V
C108 10µF 25V
C109 100nF 50V
C110 22µF 25V
C111 22µF 25V
C112 100nF 50V
C113 10µF 25V
C114 10µF 25V
C115 100nF 50V
C116 100nF 50V
C117 100nF 50V
C118 100nF 50V
C119 100nF 50V
C120 100nF 50V
C121 100nF 50V
R101 10k
R102 5.1k
R103 1M
R104 15R
R105 10k
R106 10k
R107 10k
R108 10k
R109 10k
R110 10k
R111 10k
R112 1M
R113 1M
R114 15R
R115 15R
R116 15k

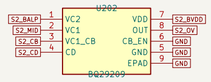
BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

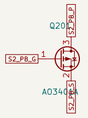


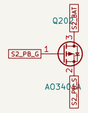






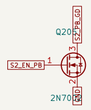










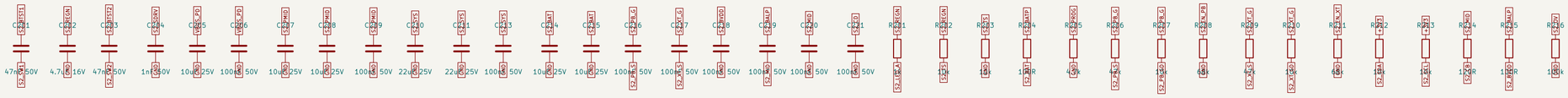












Sheet: /lipo-charger/Slot 2/
File: slot2.kicad_sch

Title: Charger slot 2: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

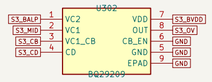
Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.8		Id: 6/14

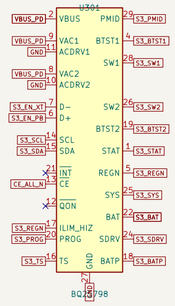
BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

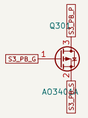












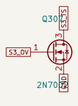






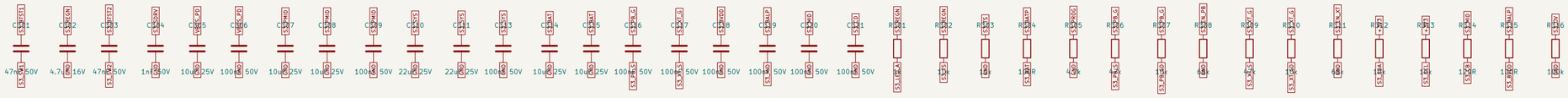












BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

J401
XT30 2S

J402
PicoBlade 1S

J403
XH balance 2S

U402
BQ29209

U401
BQ25798

Q402
AO3401A

Q403
AO3401A

Q404
AO3401A

Q405
AO3401A

Q406
2N7002

Q407
2N7002

Q408
2N7002

L401
2.2uH 3A

D401
CHARGING (green)

Title: Charger slot 4: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

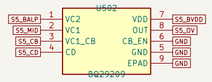
Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.8		Id: 10/14

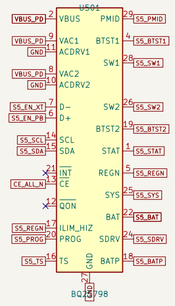
BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

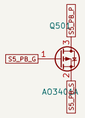


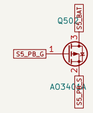


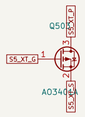


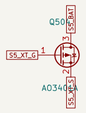












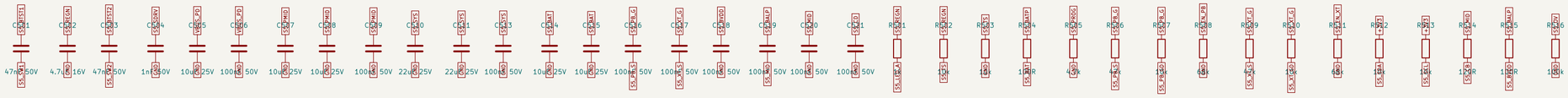












Sheet: /lipo-charger/Slot 5/
File: slot5.kicad_sch

Title: Charger slot 5: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.8		Id: 11/14

BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

J601
XT30 2S

J602
PicoBlade 1S

J603
XH balance 2S

U602
BQ29209

U601
BQ25798

Q601
AO3401A

Q602
AO3401A

Q603
AO3401A

Q604
AO3401A

Q605
2N7002

Q606
2N7002

Q607
2N7002

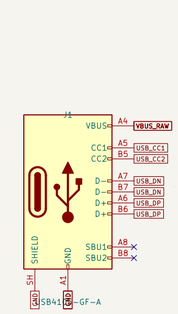
L601
2.2µH 3A

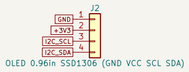
D601
CHARGING (green)

Title: Charger slot 6: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

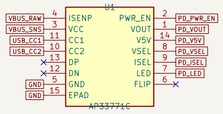
Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.8		Id: 14/14

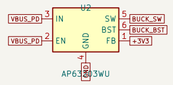
USB-C PD input (AP33771C asks for 9 V), 3.3 V buck, ESP32-C3, OLED + 3 buttons, I2C mux to the six slots.
Label-style schematic generated from design.py: pins with the same label are connected.

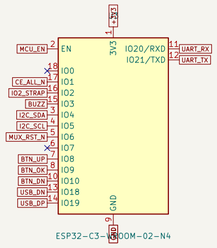






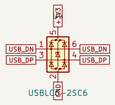


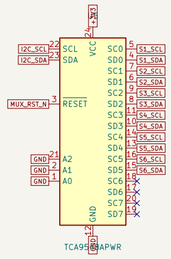










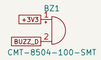






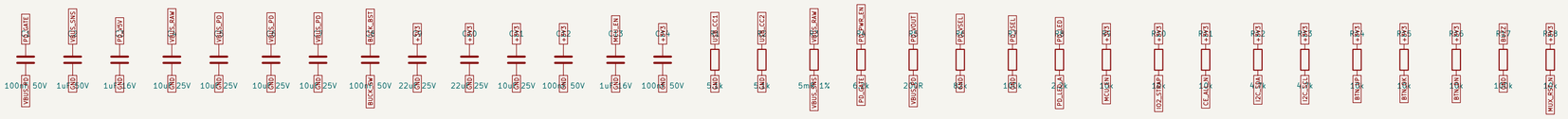










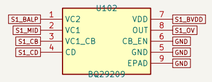

100nF 50V
1uF 50V
1uF 16V
10uF 25V
100nF 25V
10uF 25V
10uF 25V
100nF 50V
22uF 25V
22uF 25V
10uF 25V
100nF 50V
1uF 16V
100nF 50V
5mΩ 1%
2.2Ω 1%
200kΩ
100k
100k
100k
47k
10k
10k
10

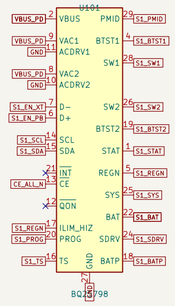
BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

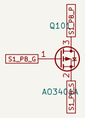


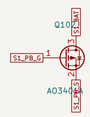






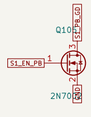




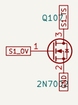






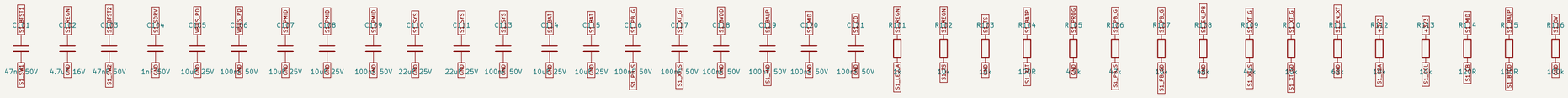










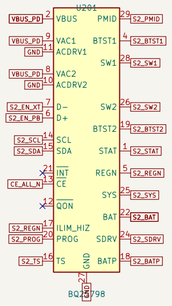


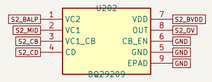
BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

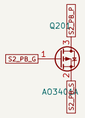


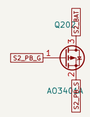






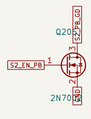




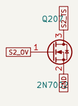






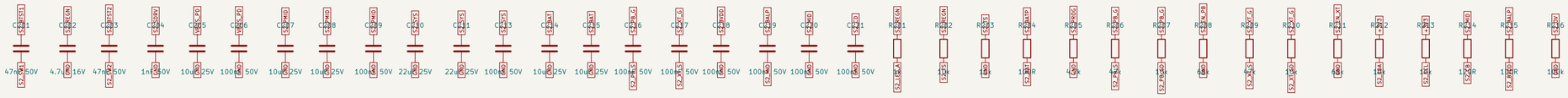












Sheet: /lipo-charger/Slot 2/
File: slot2.kicad_sch

Title: Charger slot 2: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.8		Id: 5/14

BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

J301
XT30 2S

J302
PicoBlade 1S

J303
XH balance 2S

U302
BQ29209

U301
BQ25798

Q301
AO3401A

Q302
AO3401A

Q303
AO3401A

Q304
AO3401A

Q305
2N7002

Q306
2N7002

Q307
2N7002

L301
2.2µH 3A

D301
CHARGING (green)

BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

J401
XT30 2S

J402
PicoBlade 1S

J403
XH balance 2S

U402
BQ29209

U401
BQ25798

Q402
AO3401A

Q403
AO3401A

Q404
AO3401A

Q405
AO3401A

Q406
2N7002

Q407
2N7002

Q408
2N7002

L401
2.2uH 3A

D401
CHARGING (green)

Title: Charger slot 4: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

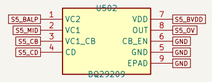
Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.8		Id: 9/14

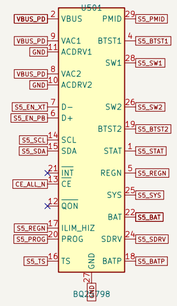
BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

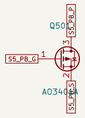








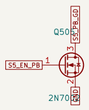


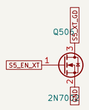








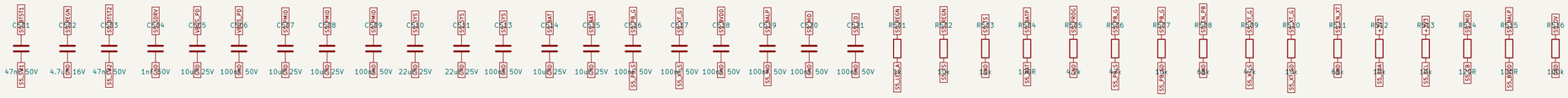












BQ25798 buck-boost charger. D+ drives the PicoBlade switch, D− the XT30 switch (register 0x47).
Each connector has a back-to-back P-FET pair so the two packs can never be connected together.

J601
XT30 2S

J602
PicoBlade 1S

J603
XH balance 2S

U602
BQ29209

U601
BQ25798

Q601
AO3401A

Q602
AO3401A

Q603
AO3401A

Q604
AO3401A

Q605
2N7002

Q606
2N7002

Q607
2N7002

L601
2.2µH 3A

D601
CHARGING (green)

Title: Charger slot 6: BQ25798 + PicoBlade (1S) / XT30 (2S) switches

Size: A2	Date:	Rev: A
KiCad E.D.A. 10.0.8		Id: 13/14